3

Justifying Automation

Most likely somebody somewhere in a company needs to justify the expense of purchasing a standard available machine or committing to the creation of a custom piece of automation. They may be the company CEO, the project manager and their boss, or some overworked bean counter in the back room. Often it may involve many if not all of them. Related to this need, the automation machine builder needs to make a profit to survive. There may be purchasing departments and sales forces involved with competitive pricing pressures pushing from every direction. How does this all fit together, and what are the different operational modes with which automation builders tend to align themselves? And who owns the intellectual property, and is getting a patent worth the effort?

These are some of the many different questions to be addressed in this chapter. The level of detail will vary by the relative importance to the automation market. Some of these questions will be sufficiently covered in brief discussions within this chapter, while the complete answers to other questions would require a set of dedicated textbooks. The level of understanding needed by an automation engineer does not have to include an MBA degree.

The topics of this chapter include:

- Traditional project cost justification for a purchase;
- Traditional costing estimating for building and selling automation;
- Win–Win purchasing philosophy;
- Maximum profit cost estimating for building and selling automation;

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- Justifying flexible automation over hard automation; and
- Intellectual property, patents, and trade secrets.

Embedded in these discussions will be other trade-offs concerning automation builders, including:

- Having one's own sales force vs. a network of independent sales representatives;
- The impact of world events due to the normal sales/purchasing cycle;
- How to deal with a single customer in a specific market vs. all of the competition;
- How does a startup automation company compete?

As we will see, there are some basic facts that assist the automation machine builder to better understand the economic games that get played. Factors like having a person blessed with natural sales abilities can be a large asset not always appreciated by an engineer. Or understanding that the lowest price does not always mean that your machine will be selected. Many purchasers dealing with competitive bids are not legally bound to select the lowest dollar amount, so reputation plays a big part in the selection process. Because the purchaser might be contracting for a machine to be delivered in six to 12 months from the date of issuing a purchase order, there is a long time of waiting that can be sleepless nights if the machine builder looks to be on shaky financial ground.

Starting an automation company has its challenges, too. It is tough to be the new machine builder on the block. Unless your new machine will make the customer millions, being the first to market, few people want to be the first to purchase your designs when economic times are slow. Everyone wants to know who else has bought a machine to minimize the risk. Getting the first few sales is challenging. The customer's warm and fuzzy feelings about the purchase deal carries more weight than one might suspect.

3.1. TRADITIONAL PROJECT COST JUSTIFICATION FOR A PURCHASE

One of the primary reasons a company will investigate automation is to reduce the number of workers. Workers are looked to be a component of a product's cost that can be adjusted. Whereas a car still needs four wheels, the number of workers used to put the four wheels on a car is not fixed in concrete (unless the union says it is). So if one worker can run around at breakneck speed and install all four wheels on a car in 20 seconds flat instead of two or four workers (this worker should probably work in a race car pit crew), the costs of producing the car will drop. Can this super worker keep it up 8 hours a day, 5 days a week? Most likely they cannot, but it sounds attractive to the financial people.

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The cost of a worker is more than his or her salary. Sometimes the internal cost to a company can reach close to twice the worker's salary. This complete financial accounting is sometimes referred to as the "burdened rate." The burdened rate may be an internal secret of a company, but one can make some guesses and be close.

Other factors can also have tremendous economic impact, but are traditionally less accepted or more difficult to quantify:

- Product quality — automation can produce a consistently good product.
- Sick or late workers — automation does not call in sick.
- Lunch/breaks — workers need breaks to survive and do their best.
- Ability to meet expanding market — automation can work a third shift on a moment's notice.

These are sometimes referred to as the more intangibles in cost justification. They are all valid points to potentially be discussed with an automation customer, but their answers may differ tremendously depending on their company culture.

Let us look at two workers in two different jobs, and see the relative attractiveness to implementing automation.

3.1.1. Fast Food Worker

The normal worker at a fast food hamburger joint may get paid something between $6 and $10 per hour. They may be stuck on some exciting job such as cooking French fries all shift, and feel like they are soaked in grease by the end of the day. Whether or not breathing in and being covered with grease is healthy for a human being (who can imaging that a doctor would suggest the job for you if given some choices), one could imagine some automation to replace this person.

Now the owners of the restaurant, besides paying the worker the hourly rate, would have to pay:

- Social Security;
- Workman's Compensation Insurance;
- Unemployment Insurance;
- Health Insurance;
- Other assorted payroll taxes and fees.

So the hourly rate to the owners is effectively $3 to $4 per hour greater than what the worker sees in their paycheck (before the personal withholdings). So the total burdened hourly rate is now $9 to $14 per hour. What kind of automation would that start to justify?

The next set of questions to the owners is:

- How many shifts will the worker be replaced?
- Does their French fry person do other tasks?

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- How many months or years is the owner willing to invest before seeing a return?
- What is the risk factor dealing with the automation being successful?

For the hamburger joint, most likely it is 2 shifts, 7 days a week. So one could take the burdened hourly rate (assume an average rate of $12 per hour) and multiply it by 16 hours per day times the 7 days a week. That produces an amount of $1344 per week being effectively paid out for the French fries employee.

If the owners were willing to invest the amount such that after one year's time the automation has been paid for, the total for the automation would be almost $70,000. (A one year payback period is common in industry, but it can range from 90 days for short-term markets, to two or three years for stable long-term markets.) Added to this are the intangibles, where workers cooking French fries are not always excited about their work, and some are likely to be late or call in sick. The process is well defined, so product quality is not a huge variable.

The author thinks that it might be possible to build and sell automation to this industry at this price, assuming that dozens if not hundreds or thousands of restaurants purchase copies of these machines, so as to spread out engineering design costs and to obtain cost savings by making multiple automation machines in batches. To carryout this scenario, many of today's chains of hamburger joints have many different store configurations, and they may have to do some standardization to make it work at this automation production point.

But do the owners have $70,000 to purchase such a machine at any specific time? Will they have to take out a loan? Can they lease the machine from a leasing firm? Is their credit good? Are they financially strapped as it is? Not knowing how the hamburger chain's finances are positioned, one is left to make an educated guess. The author's guess with his limited knowledge of their internal financing is "maybe."

3.1.2 U.S. Postal Worker

The author has had multiple interactions with the United States Postal Service (USPS) over the last six years. It has been told to him by USPS engineering managers that the internal justification amount per employee to use automation is over $150,000 (J. Weller, personal communication, 1996). This means that the automation to replace an employee can cost $150,000, and with whatever internal accounting process they have, everyone is happy. Many of the USPS jobs are a single shift, with a late afternoon start through very early morning hours. Some of the driving components to this cost are high salaries, fantastic retirement plans, and great health benefits.

Many of the human tasks in a USPS processing center are where people are feeding odd shape envelopes and/or bulk mail (magazines) into current

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automation sortation machines. The sortation machines do a great job considering the range and quality of mail that is handled every day, but the human operator is basically a slave to the machine, keeping it well fed. The operators risk repetitive motion injuries like carpal tunnel, and many workers there do get long-term disability care that costs the USPS a great amount. Thus the great health benefits are needed and used often.

The $150,000 amount sounds attractive to an automation builder, but the USPS must be in a mode that is financially viable to create such a large purchasing program. There are around 1000 processing centers in the United States, and for the worker replacement program to be cost-effective, a good number of the 1000 sites must be upgraded in a short amount of time. So when the USPS is short on cash as they are in the early 2000s, things move slowly.

3.1.3. Which Market to Approach?

As we have seen, the USPS market seems to be more attractive due to the cost justification numbers of $150,000 vs. the French fries automation for $70,000, but the job of automating French fries might be easier and require less machine complexity and component cost. However, the USPS market would be a single huge contract that might be a windfall if one gets selected, or a massive disappointment if not.

The only suggestion is to work on solutions to both markets on one's free time, and approach them both when given the opportunity. So when either customer can come up with the right amount of dollars to go forward, one is ready to play the competition game.

3.2. TRADITIONAL COSTING ESTIMATING FOR BUILDING AND SELLING AUTOMATION

Developing a good cost estimate for building and selling a new piece of automation is a combination of good financial practices, experiences from past projects, and knowing you have a good method of manufacturing operations that will follow the estimates fairly well. You plan for a slight cost overrun in the form of a contingency, and you plan on making a profit. To understand the core issues here, we will look at a more reader-friendly example.

3.2.1. Room Refinishing Example

Because many readers of this text may not have any experiences with costing out an automation project, let us first focus on a more commonly found problem: refinishing a room in a home or apartment. Let us assume that the goals are to:

- Replace the floor covering;
- Repaint the walls and ceiling;

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- Replace the ceiling light fixture;
- Add a pair of bookshelves to one wall.

These do not look like overwhelming efforts on first glance. Assuming one has the average person's skill set, or a friend or two who do, the tasks do not look too daunting. A paint sale in the morning paper makes the project seem timely, and you always wanted to get better lighting, so you can cost-justify replacing the old light fixture. The room will be more useful with that nice set of bookshelves, too.

So how long will it take? How much will it cost? What if any expertise will you have to hire out? These are questions that usually are not completely answered by many homeowners or apartment dwellers before they start a project. You see a price for paint at $20 per gallon (you will use maybe three gallons?), carpet for $240, and bookshelves for $100, and you guess the light fixture will be $75. You have a quick total of $475 and your bank account says you could afford $500, so you go ahead with the job. You think it might take you two consecutive weekends if all goes well. Assume four killer days of 10 hours each for a total of 40 hours. (Some people's spouses automatically double their mate's estimates.) Let us see what the impact of a quick but incomplete analysis might be.

Floor Covering

You have an old carpet that is well worn out, so it goes to the trash. But your trash hauler charges you $20 to take it away. The new carpet is $15 per square yard, and your 12 ft by 12 ft room requires 16 square yards. But you would like it to be a wall-to-wall installation, and you need 48 ft of the thin wood pieces that have the embedded nails at the correct angle to hold it in place, another $40 (Fig. 3.1). You might want it installed professionally to remove any wrinkles, but we will go on a limb and say that you have the right friend with the right tool and they owe you a favor.

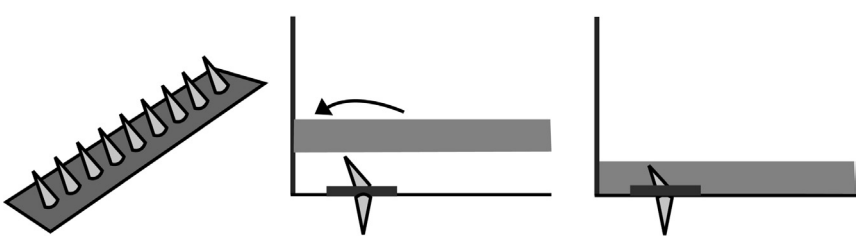


FIG. 3.1 Wall-to-wall carpet strips

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Painting the Walls and Ceiling

The walls require two coats of paint so you get two gallons at $20 each. The ceiling can be done with one gallon at $20. But you need at least one paintbrush, one roller and roller pan at $20. You do have a stepladder, so no need to purchase one, but there are some nail holes that need patching and sanding first. So the patch material, putty knife, and sand paper cost you another $15 you were not planning to spend.

New Light Fixture

For some reason unknown to the author, light fixtures always seem to cost about twice the amount you plan for. Maybe it is the custom use of glass that makes even a moderate fixture somewhat expensive. So the initial guess was off, and you spend $150. We will say that you can rewire the light without giving yourself an electrical shock by the proper selection of the circuit breaker to turn off and good common sense, but you do not have a roll of electrical tape and connector nuts, so you spend $5 more. The old light fixture can hopefully be thrown out in normal trash after some effort to dismantle it.

Adding Pair of Bookshelves

You purchase the bookshelves at a discount home improvement store for $100. They are nothing fancy, but you are a practical engineer. You would like to find the studs in the wall so the shelves can handle all of the textbooks you have accumulated from your school days, or you might have to settle for some of the wall anchors that came with the shelves. The wall studs might be too hard to find, or they are located in poor locations for good room layout efficiency. The wall anchors look adequate to do the job, but do not give you the sense that they would handle all of your books. So you buy stronger wall anchors for an additional $20 and you are more content (Fig. 3.2).

Total Costs and Time Required

Remember your initial cost estimate was $475, and this amount was within your budget, so it energized you to do the project. Also remember that you and your friends' labor costs are free (except that you probably will have to feed them a few times), but if we focus only on hard purchase costs and the normal underestimates and surprises, we find that we have spent $665, an additional $190 or 40% overrun. This does not include the cost of food and beverage to keep the "free" workforce happy.

So you have to find the additional $190 from your personal budget. If this happens once every two years or so, you look upon your refinished room with pleasure and hopefully just make do. You say you will try harder next time to have a more realistic budget. You convince yourself that the free labor was worth far more than the additional $190 and you move on to something else.

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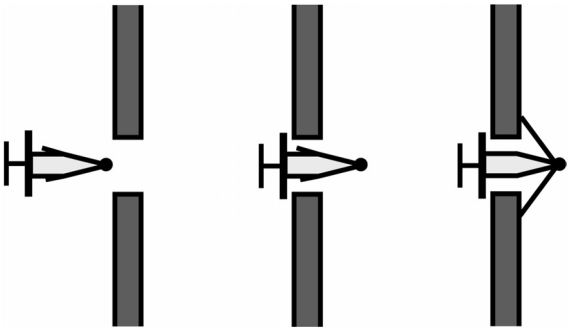


FIG. 3.2 Superior wall anchors

The time estimate of two weekends was not even addressed in the previous discussion. Can or will it be done only on two weekends and no additional evenings? It is harder to estimate how much talent you and your friends have here. But let us assume that you spend a few evening hours making purchases, getting things prepared for or cleaned up from the prime weekend time, for an additional 10 hours. This is a time overrun of 25%, but of no real consequence to a homeowner.

3.2.2. Impact on a Machine Builder

If you were running an automation building company, and you constantly have material cost overruns of 40% and time overruns of 25% of your labor, you would be out of business very quickly. When all of the accounting is done on a successful project, the machine builder might make a profit of 10–12%, so there is little room to absorb errors of this magnitude.

One might argue that the room refinishing project is the kind of job you do so infrequently you forget the lessons learned from the last time. If you did remember these lessons, you might look at your budget and decide to wait a few more months until you save more money. But the impact of your room decision is probably not going to make you bankrupt.

A company that makes all of its sales by making proposals and giving budget costs it must later work within must make better estimating efforts. We can learn from the example, however, where many of the estimating mistakes can occur. We can:

- Be swayed by the advertised price and not consider hidden costs (trash hauling);
- Use data from memory that is not very accurate (light fixture);

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- Not perform enough critical calculations (wall anchor needs);
- Not define the customer specifications completely (carpet being wall to wall install);
- Underestimate processes (wall preparation before painting);
- Underestimate time to do all of the tasks (shopping, preparations, cleaning).

So our quick little example manages to hit many of the key issues project costing will grapple with, whether automation or most any project. One needs to know the project specifications and requirements involved, and to minimize the number of surprises.

3.2.3. Cost Structure for an Automation Builder

An automation machine builder may have a line of standard available products, or they may design and build custom machines, or they may do both. Reputation and word of mouth can help to initiate the sales process, but this is usually not enough. So there is a need for an internal sales force and/or a network of external sales representatives. External reps are usually independent agents who work for a commission. These commissions are paid when the sales are complete, and are in the 5–25% range, usually 10% for automation machines.

External reps may seem at first glance to be the best way to approach sales. They cost you relatively little, and you only pay them the commission when sales occur, but these reps are most likely handling a range of other products and automation machines. When they walk into the door of a potential customer, they have a large three-ring binder filled with product literature that includes your machine and maybe 10–30 other machines made by other manufacturers. So they are listening to the customer, and looking for the easiest sale combined with the biggest commission check. Some reps will be great for you, others disappointing. It takes time to select and train a good network of reps that will potentially cover the entire United States and beyond.

Cost Components Within an Automation Project

Many of the components of an automation project will probably roll off your tongue. Others are less obvious. Some can be allocated to a specific project, others are for the well-being of the company and no-one really knows to which account to charge it. Good management practice just knows it needs to be done sometime somehow. The following is a list of expenses when one deals with automation:

- Purchased components;
- Engineering design time;
- Fabrication time;
- Inventory and Work In Progress (WIP) financing;
- Profit (10–15% hopefully);

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- Administrative costs;
- Fringe benefits;
- Workers compensation insurance;
- Unemployment insurance;
- Travel to see the customer before and after the sale;
- Installation costs (depending on contract agreement);
- Warrantee costs (depending on contract agreement);
- Proposal writing;
- Commissions to product reps if used;
- General marketing expenses (trade shows, literature, websites);
- Real estate related overhead;
- Contingency funds.

Some of these are worthy of note. Warrantee costs are for the normal one year parts and usually labor expenses, which might run to 3–7% depending on the machine. Travel to see the customer and write the proposal need to be included, and might appear as an administrative cost, since customers do not want to think that they are or have paid for the initial contract development work. However, it does need to get paid for from some account, and if one ignores this for multiple projects, it will eat up the profits.

The contingency funds approach to automation development is sometimes used when there are uncertain issues or true development needed, since this is a novel or one-of-a-kind machine. When both the customer and the builder acknowledge the risk factors, a range of costs needs to be covered in the proposal if things do not go optimally. So as to not create a proposal with a lot of excess built in, a contingency amount is specified. This can be handled in several ways. One way is to state that the amount (perhaps 10–20% of the total amount) will not be used without the customer signing off at an agreed-upon milestone. Or the amount might be approved, but the builder agrees to provide documentation to prove the need, or is compelled to give a rebate at the project's conclusion.

Groups Involved in Automation Development and Payments Timing

There are many small and larger variations to the norm, but here is one possible scenario on how automation is developed and paid for:

1. The internal sales force and/or external sales reps bring in a Request For Proposal (RFP).
2. The engineering and management teams use experience and financial data to make a step-by-step assessment and develop the proposal.
3. The proposal is submitted to the potential customer by the sales force or reps.

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4. The proposal is hopefully selected from competition: 5–30% success depending on market, customer type, and the purchase cycle timeframe.
5. Depending on the customer's budget cycle, the selected proposal may get approval in a few days or a few months. Most proposals have an "effective until" date.
6. The customer issues a Purchase Order and may be required to place a down payment (20–30%, negotiable).
7. The management team dusts off the proposal and implements the project timeline.
8. Engineering design and development produces CAD drawings. They generate a list of off-the-shelf components and subcontracts for custom parts for the Purchasing Department.
9. Manufacturing produces any in-house components.
10. Management determines timing for purchases deliveries and internal production and schedules assembly. It notifies the customer if some component lead time will impact delivery.
11. Assembly builds the machine.
12. Engineering works on the debugging procedure.
13. Optional Progress Payment may be required depending on contract language (20–30%, negotiable).
14. Quality Assurance meanwhile inspects internally made components and any critical purchased components where needed.
15. The customer comes to visit the machine for a performance review.
16. Major Payment is made, bringing things to the 85–90% paid level.
17. The approved machine is shipped to the customer site and installed.
18. A final check is issued (10%) when the customer is happy or begrudgingly agrees to pay.

There are often changes to the automation machine design because this is usually such a long process (4–12 months). A machine in early testing might show some performance that was originally thought to be great, but is now no longer desired. Or the product to be assembled is modified. Authorization needs to be given to deviate from the original specifications. So an Engineering Change Order (ECO) is issued, and depending on who issued it and why, some additional payment may be required. During the first development of a machine, whether it is $50,000 or a multimillion dollar line, if the change orders from the customer comprise less than 10%, things are proceeding normally.

3.3. WIN–WIN PURCHASING PHILOSOPHY

What is another approach one could take towards costing and sales? Instead of just using cost plus 10%, one could know what the project is worth to the

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customer. If the customer shares enough information with the machine builder, they can work together to see what are the most important needs to be addressed by each side, and recognize that each side should benefit. This has been referred to as a "Win–Win" situation. Here, the customer states how they will save money by replacing labor and reducing the other less tangible costs. The machine builder shares enough about his pricing and component costs that both sides can agree on a price where a profit can be made but the customer does not feel that they are being gouged.

This is a wonderful situation when it can occur. It does rely on both sides being open, and this sometimes can be quite a game. One of the required pieces of information is how many people will the automation be replacing. How much does the customer's accounting method attribute to each person being replaced? What are the current Workman's Compensation expenses for each worker? What timeframe does the company require for a project to be effectively paid back?

Answers to these questions can easily be put into a spreadsheet to determine the potential savings. Then the automation builder can determine what the value is to the customer. If everyone is up front, a price point where both sides make or save money is the "Win–Win" situation. The builder will prosper enough that they will survive and be able to support the currently purchased machine for years, make additional machines for future sales, and all is good in the world.

However, the games can begin if the customer wants to keep more of the potential savings. They may "stretch the truth" about how many workers currently are performing the job. They may not be forthcoming about a second or third shift. Finding the correct numbers to enter into the spreadsheet can be difficult. The author has seen this happen several times personally. A true "Win–Win" needs more honesty than this. It needs bilateral trust.

3.4. MAXIMUM PROFIT COST ESTIMATING FOR BUILDING AND SELLING AUTOMATION

Some automation builder companies are aggressive towards every dollar they can get. They are not concerned about "Win–Win" as much as growing their balance sheet. They know how many workers their machine will replace. They know they have the only solution available. They have filed for a provisional patent on their concept and will aggressively defend it. They know that having the leg up on the competition does not last for decades, so one must go for the brass ring with every sale.

So, with this strategy, the price for the automation is just short of the maximum that the customer will be able to justify. They know that there will be repeat sales, since they have their customers over a barrel. The first machine designed

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and built costs more than the second and third one. Then who cares if building a second machine is only $15,000 when the repeat customer can cost justify it for $65,000. The extra $50,000 is all profit and the machine builder wins big time! But some customers will begin to realize the game that is being played here, and sometimes will try to find other sources to make future copies of your machine.

Another clever strategy that may not seem as outwardly aggressive can be found in the packaging area. A popular new product package has a strong set of international patents for protection. The automation machine and the required metal backed paper used to make the product are both covered by these patents. The machine builder will lease the machine for something unbelievable like $1 a day, but requires users to buy their special paper at an outlandish price. And because no-one else can supply this special paper, they have a monopoly. World acceptance of the packaged product helps to drive the monopoly towards great profits, yet their machines are seen as cheap. This ends up being extremely expensive once the total costs are calculated.

These profits are seen as so great that other forms of competing packaging are being thrust upon the consumers. Some potential automation packaging customers refuse to buckle to the high costs of the monopoly. The new competing product is now making headway, but the owners of the monopoly are still in a good financial situation.

So these aggressive companies, following no matter which strategy, are looking to make as much money as possible, for as long as it lasts. In their defense, the consumer market is so flighty at times that one cannot blame them. They may be out of business in 1–5 years if they do not expand their product line, but that is not the main goal — making money is.

3.5. JUSTIFYING FLEXIBLE AUTOMATION OVER HARD AUTOMATION

Many of the machines described in this text can be classified as "hard" automation; that is, they cannot be easily modified and reprogrammed to do another task. A hard automation machine that assembles car stereo faceplates will not be able to assemble a car generator. Hopefully the hard automation has been optimized to crank out the product faster than what is referred to as "flexible" automation. Flexible automation usually means that there are one or more industrial robot arms as the key elements in the workcell.

As one can imagine, there are financial scenarios when either type of automation can shine over the other. Part of this can be seen as the reuse of a significant part of the automation cell vs. the option of creating a potential boat anchor when one creates a single-task machine, but if that single task is to be performed

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for many years (such as the assembly of spark plugs), then the flexible automation may or may not be able to keep pace.

Brian Carlisle, CEO of Adept Technologies, presented some business research (Carlisle, 1999) he conducted in the late 1990s. Adept is a leading U.S. manufacturer of assembly robots, so it was in his best interest to show how flexible automation can be the best choice. He also compares automation to manual production, but raises some good points no matter which side one is trying to take. Figures 3.3–3.6 shows some of his findings dealing with financial justification. His key observations are:

- Most automation engineers do not have a readily available financial comparison tool;
- Quantifying flexibility had not been done before his research;
- Depreciation was often used as a hindrance factor by accountants, but does not have much impact;
- Rarely can a company make up financially for lost time if changeover takes days or weeks.

Potential customers have to know how a company carries out internal finances, what it wants out of automation, and what the range of possible products to be manufactured will be in the next few years. Without this information, it is tough to justify anything.

3.6. INTELLECTUAL PROPERTY, PATENTS, AND TRADE SECRETS

Intellectual property is the term used for the collective group of patents, trademarks, copyrights, and trade secrets. Depending on a company's size and sales volume and the market conditions, intellectual property can be one of the strongest assets a company can own. These assets can be created, purchased, or sold. They are the benchmark of many startup companies and their investor funding potential. They can also keep a room full of lawyers employed for decades. Many times when an employee starts work at a company, they are required to sign away any or all of their intellectual property rights. Check with your

- Explain Important Financial Metrics
- Provide Tool to Automation Engineers
- Compare Flexible and Hard Automation to Manual
- Help Sell Proposal

FIG. 3.3 Purpose of justification analysis (from Carlisle, 1999)

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- Quantify Value of Flexibility
- Changeover Cost Analysis
- Incremental Capacity
- Show Relative Sensitivity of Key Assumptions

FIG. 3.4 Generate justification automatically (from Carlisle, 1999)

company if you are currently employed. Or if you have been employed, your past company may hold certain rights for some time after you leave.

Each of these terms will be briefly discussed, with focus on what is common to automation companies and their customers.

3.6.1. Patents

The granting of a patent from the United States Patent and Trademark Office (USPTO) can be one of your career's highlights. It means that you have invented something new and novel that no one else has thought about before. It gives the assignee (you or your company) specific rights in ownership of the idea in bringing the concept to market. If some other person or company tries to use your idea in the protected timeframe, you can go after them in a court of law. There are a significant number of rules that will not be addressed here, so the reader is referred to the USPTO website (www.uspto.gov).

The key point in an invention is that it must be something that is not currently available in the market, that has not been patented before, and is not an obvious extension of these existing works to one who is a practicing engineer in the field. This means if someone takes a good current product and simply paints it red, one cannot get patent.

- Changeover Scenario
- Changeover Times and Ramp-up Rates
- Is Revenue Lost During Changeover Recoverable?
- Special Packaging for Automation
- Cost of Quality
- Financial Impact of Errors
- Cost of Labor
- Utilization

FIG. 3.5 Most important assumptions (from Carlisle, 1999)

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- Depreciation Life!!
- Depends on Profits and Tax Rate
- Cost of Utilities
- Cost of Floor Space
- Cost of Maintenance

FIG. 3.6 Least important assumptions (from Carlisle, 1999)

Patents do not cost a great amount strictly from the USPTO fees structure, but to properly interface with the USPTO office and to fulfill all of the regulations and rules on style and types of drawings, etc., one often relies on a patent attorney. This normally brings the cost of getting a U.S. patent to something between $5000 and $20,000 on average. International patents can range from $3000 to $20,000 for each country in which one thinks that there is a viable market. So if one foolishly thinks that a new patent on space heaters will sell well in the Amazon region, then one can waste the dollars getting a patent there.

However, for every U.S. patent issued, perhaps only 5–10% of them actually generate any income. The patent process takes time, energy, and money. Inventors always want each of their inventions to be patented for a chance to get rich, but the market is fickle, and other people come up with a better method, a better machine, and therefore a better patent before one has recouped the costs of one's own patent. There are a tremendous number of great ideas one can see by searching the USPO website database. It is free of charge. And it will tell you if a patent has expired or has been abandoned. It will not tell you if the idea ever came to market, however.

Patents that are issued have a renewal fee every three to four years. So if a patent is not profitable, it may be left to expire by the inventor or his company. This leaves it open to the general public to use free of royalties or other financial obligations. It makes for interesting surfing of the net to see what useful and crazy machines have been designed, but possible never built! One does not have to build a machine to get a patent on it.

3.6.2. Trademarks

Trademarks are the name or term that a product is marketed under. It helps tremendously to establish brand recognition, and to assure repeat customers. It is something that everyone needs to respect so as to not violate the trademark laws, but this is something usually left to the marketing department. At the level most automation engineers work at and conduct their business with customers, there is little to be concerned about trademarks. That is the job of Marketing.

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3.6.3. Copyrights

Copyrights have to do with written documents, music, photos, and similar products. On the surface, they would seem to have little to do with automation, except when one gets to the point to write the user manual for the machine one has built. However, there are several tricks sometimes used that have not been necessarily proven to be useful, at least not yet.

Some companies will copyright the proposals they submit to a potential customer for protection. It is not unheard of that a shady customer likes the concept given in a proposal, but does not like the pricing. The customer might try to shop around the proposal concept to other manufacturers to see if someone else can make the same machine for a lower price. Or the customer decides to make the proposed machine internally. Both of these practices are very unethical, but unfortunately not unusual. It is common enough to make proposal writers very anxious. So the proposal is marked with a "Copyright" designation, in the hopes that if some other entity tries to use this proposal concept, it would be infringing. This approach seems to have limited legal effect other than to possibly scare away the potential new manufacturer, who might have been duped into trying to quote someone else's concepts.

3.6.4. Trade Secrets

When a company has a machine that represents a large commitment of time and dollars, and it is believed to be vastly superior to commercially available machines, that company may not try to patent it at all. It will just keep it as a trade secret. That is because one must present all of the technology in the patent disclosure, and it is then available for the competition to review and perhaps improve enough to warrant a new patent. With Trade Secret policies, employees are bound by agreements not to disclose the technology under penalty. Visitors are not allowed to see the machine. No-one takes photographs. It is simply a well-kept secret.

This strategy is most likely less expensive than obtaining and defending U.S. and international patents. One small company known to the author budgets $50,000 per year to move a patent along in the international process (which takes more time than the U.S. process) and to defend the patent when it is awarded. This is for a single invention! Some people say a patent has not really proven itself until it has been defended from another party in court, and patent litigation is very expensive.

3.6.5. Intellectual Property for Automation

With all of the expense of obtaining and defending a patent, it may not be the right thing for all machines. First, if it is in an area that is very mature, the core

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patents may have expired 20–30 years ago. To make a better machine means you have to try some radical new way to do it. Secondly, one has to generate enough profit to pay for the patent work. This most likely will not happen if you are making a single special purpose machine, and therefore unable to spread out your engineering design and development costs. So if you have a product that your automation building company is developing, and you plan on selling dozens to hundreds of them, then it may be fruitful to proceed with a patent.

In today's climate for contracts and business negotiations, often the customer who is contracting for a piece of custom automation will want the intellectual property, including the patent rights. Some customers might let you retain the patent rights if they get their machine for a cheaper price, but this is usually in a situation where if their competition gets a copy of the same machine, the first customer feels like they will still have the superior product. Or they may allow you to sell into markets that do not compete with their defined product line.

3.7. CONCLUSIONS

Whether the current automation challenge before you is the first or the 100th machine to design, one should look at all of the costing components in Sec. 3.2.3. Many of these 18 steps will become obvious after a few times down this road, but if one changes employment between the previous automation design and the current project, a designer can be surprised by how many of these steps can be different due to company policies and culture.

When it comes time to plug some numbers into these categories, it is always best to have someone else review the suggested numerical entries. Even on your 100th machine you may forget to include some sensor or actuator that just seems to have slipped your attention. This process of costing a project can turn a fun and potentially profitable time into a nightmare that seems to last for years!

PROBLEMS

1. Create a cost and time estimate for:
   a) A five-course dinner.
   b) A party for 10–12 people.
   c) Refinishing a room.
   d) Some other event.

   Keep notes on costs, time, and unfulfilled expectations. Look at your estimating time as a function of the total time. How did you handle any cost overruns?

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2. Search the USPTO website database (www.uspto.gov) and try to find a machine that:
   a) Mows the lawn.
   b) Washes the dishes.
   c) Does laundry.
   Which of these correspond to consumer products than you have seen? How do these differ from commercial products?
3. Types of sales forces:
   a) How do you deal with commission-based sales people, like car dealers or some large appliance dealers?
   b) How do you react with full-time sales staff, such as a waiter in a restaurant?
   c) Who does the better job and why in your opinion?
4. You have just invented a new automation machine that everyone will need in their home. It peels and slices fruit with no human intervention. What sales method would you use and why?
   a) Dedicated sales force going door to door.
   b) Commission-based sales force in shopping mall kiosks.
   c) Phone operators and television infomercials.
   d) Internet-based sales provider.
   What advertising would you need to support your selected sales method? What would make the everyday homeowner want to part with $100 for your device?

PROJECT ASSIGNMENT

Using one of the projects from the Appendix (or any other projects), perform the following:

1. Estimate the current human operator cost of processing the product. Use the following labor rates (These include estimated company overhead rates, for unemployment insurance, worker's compensation, social security, etc. These are not the take-home pay rates of the fictitious employee):

Food Service	$10.00/hr;
Light Assembly	$14.00/hr;
Machine Operator	$20.00/hr.

   To do this assignment, you most likely do not have enough hard facts to be totally accurate. Perform a mock processing of the tasks and have someone record the time needed.

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2. Create a list of probable reasons why one should automate this process. Concerns should include the relative likelihood that manually there would be:

   Improvement of quality;
   Reduction of repetitive motion injuries;
   Improved productivity;
   Ability to meet expanding market.

3. By looking at the probable steps you developed in Chapter 2, make an educated guess on what the parts and labor will be to make such a machine. There is not sufficient information given in this text in Chapters 2 and 3 so far to do this accurately, but view this task as an interesting estimate that you will look at again after you are done with the total design. It is interesting to see how sometimes one can be fairly close to the actual cost with limited knowledge, and yet other times are off by an order of magnitude.

REFERENCE

Carlisle, B. (1999). Financial justification analysis for flexible automation. Adept Technology System Integrators Conference. Monterey, CA, January.

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